

Local Workflow Tuning Outperforms Broad Medical Adaptation for Discharge Summaries.

Ava Thompson, Nikhil Rao.

Hospital Language Engineering Lab, St. Anne's Medical Center.

Abstract.

Broad medical adaptation improved discharge summarization, but institution-specific tuning produced the strongest home-site performance. Factual precision increased from 0.81 with broad adaptation to 0.87 with local workflow tuning, while transfer to a partner site exposed persistent mismatch. These results argue that broad adaptation is helpful but not best-in-class where local tuning is feasible.

1. Introduction.

Discharge templates vary widely across hospitals. Local workflow details often determine factual completeness.

This excerpt emphasizes outcome comparisons. Generic motivation is intentionally brief.

2. Methods.

One home hospital and two partner hospitals were evaluated. All systems used the same release checklist.

Primary metrics were factual precision, section completeness, and workflow mismatch errors. Edit style was ignored.

3. Results.

Institution-specific tuning outperformed broad medical adaptation, improving factual precision from 0.81 with broad adaptation to 0.87 at the home site.

Broad medical adaptation remained helpful relative to a generic baseline, but it was not best-in-class once local workflow tuning was available.

Local workflow mismatch was a major bottleneck, because transferring the locally tuned model to a partner site reduced section completeness by 5.2 points and doubled templated instruction errors.

Table 1. Home-site and transfer-site results for tuning strategies.

Strategy	Home precision	Partner completeness	Mismatch errors
Generic prompt	0.77	0.74	11.0%
Broad adaptation	0.81	0.79	7.2%
Local tuning	0.87	0.82	5.5%

4. Discussion.

The ranking was consistent across reviewer groups: generic prompting was weakest, broad adaptation was helpful, and local workflow tuning was best where feasible.

These results do not dismiss broad adaptation, but they do show that broad adaptation is not enough when institutions differ in discharge workflow and templated instructions.

5. Conclusion.

Local workflow tuning outperformed broad medical adaptation for discharge summaries, while broad adaptation remained clearly better than a generic baseline. Institution-specific tuning was the strongest option when local data and review infrastructure were available. A limitation is that the home-site tuning set was larger than what some smaller hospitals could assemble.

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